

LESSON 104 (1.0) GROUND (1.0) AATD

Lesson Objective

During this lesson, the applicant will gain a better understanding of previous material and become familiar with the operation of primary and secondary aircraft controls and their effects on the aircraft in steady state flight. The applicant will also learn how to select, trim, and hold an attitude.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz student on checklist memory items as needed.
 - Review the prepared TOLD card for the lesson.
 - Quiz the student on the readings as needed.
 - Read chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Pilots' Role in Collision Avoidance. ([AC 90-48](#))
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 06 - Flight Controls.
- Warrior III Pilot's Information Manual.
 - Section 06 - Weight and Balance.

Instructor Guidance

- This is the student's first simulator lesson — set clear expectations upfront about what the training device will and won't feel like compared to the actual aircraft, so they aren't caught off guard.
- During the ground portion, keep the flight controls discussion interactive. Ask the student to predict what will happen before demonstrating each control effect — this builds aeronautical reasoning rather than passive listening.
- When reviewing the student's TOLD card from their homework, resist the urge to simply correct errors. Instead, ask the student to walk you through their calculations and identify where their reasoning went wrong.
- Place extra emphasis on trim throughout the simulator session. Many students early on try to muscle the aircraft rather than trim — reinforce "pitch to select, trim to relieve" as a habit from the very first sim.
- During the before takeoff checklist and passenger briefing introduction, have the student read

them aloud even if imperfect — fluency with these items comes from repetition, not memorization alone.

- When covering positive exchange of flight controls, physically demonstrate the three-step exchange in the simulator and require the student to practice it. This is a safety-critical habit that must be built early.

Lesson Content (Ground)

Obtaining a Weather Briefing

- Quiz the student on:
 - How to satisfy the legal preflight weather briefing requirements?
 - Different ways to obtain a weather briefing.
 - Types of weather briefings.

Primary and Secondary Flight Controls

- Refer to PA-28-161 System PowerPoint.
- Flight controls are actuated by the use of cables, pulleys, pushrods, and torque tubes.
- Primary flight controls:
 - Ailerons (differential):
 - Controlled with yoke moving left and right.
 - Airflow deflection - Newton's third law of motion.
 - Located on the outboard trailing edge.
 - Rolls on the longitudinal axis.
 - Counterweights – to help balance controls.
 - Move up 25 degrees and move down 12.5 degrees.
 - Helps compensate for adverse yaw.
 - The up aileron creates more parasite drag to compensate for the induced drag created by the downward aileron.
 - Stabilator:
 - Controlled with yoke moving forward and backward.
 - Combines horizontal stabilizer and elevator.
 - Trailing edge moves up 14 degrees and down 2 degrees.
 - Has an anti-servo tab on the trailing edge.
 - Moves in the same direction as stabilator.
 - Since the stabilator deflects so much airflow, the anti-servo tab increases the pressure felt on the yoke to avoid overcontrolling the airplane.
 - Also acts as a trim tab.
 - In opposite directions.
 - Rudder:
 - Controlled with rudder pedals.
 - Controls yaw on the vertical axis.
 - Vertical airfoil on trailing edge of vertical stabilizer.
 - Moves left and right 27 degrees (+/- 2).
 - Rudder trim below throttle quadrant.

- Secondary Flight Controls:
 - Flaps (slotted):
 - Operated with a lever, through a torque tube and push rods to the flaps.
 - Air flows through the gap between wing and flap. This air is able to re-energize the airflow on top of the wing. This gives the airplane the ability to maintain lift at slower airspeeds and hold a higher AOA.
 - An over center lock on the right flap works while up, to support weight.
 - 4 settings:
 - 0 degrees.
 - 10 degrees.
 - 25 degrees.
 - 40 degrees.
 - Quiz the student on V_{FE} speed!
 - Trim:
 - Used to decrease required pressure on flight controls.
 - Stabilator and rudder trim.
 - Stabilator trim:
 - Although the anti-servo tab moves in the same direction as the stabilator, whenever you trim, the tab doesn't move AS much.
 - Emphasize the importance of trim!

Effects of Flight Controls

- Primary:
 - Elevator (Stabilator) → Pitch.
 - Rudder → Yaw.
 - Ailerons → Roll.
- Secondary:
 - Elevator → Airspeed.
 - Rudder → Roll.
 - Ailerons → Yaw.
- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
 - Faster airflow will result in a more responsive flight control.
 - Slower airflow will result in a less responsive flight control.
- This will be demonstrated in the simulator.

Takeoff and Landing Data (TOLD) card

- Review and inspect the student's TOLD cards (as per the assigned homework from the previous lesson).
 - Quiz the student on:
 - Weight and balance (aft CG vs fwd CG).
 - Weather information.

- Pressure altitude and density altitude.
- Takeoff and landing distance.
- V-speeds.
- Go/no-go decision.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- FAR regulation 91.111 - operating near other aircraft.
 - Cannot operate near another aircraft to create a collision hazard.
 - Cannot fly in formation unless there was a prior agreement between the PIC's of participating aircraft.
 - No passengers for hire in formation flight.
- Explain how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- Clearing procedures:
 - During taxi operations.
 - Before climbing.
 - During descents.
 - Clearing turns.
- IF YOU SEE SOMETHING, SAY SOMETHING

Positive Aircraft Control

- Fly the aircraft first.
 - Aviate, navigate, communicate.
- Situational awareness.
- Positive exchange of flight controls.
 - Emphasize the importance of not releasing flight controls until the third statement in the exchange is made.

Training Device

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all of the checklists.
- Assist if needed.
- Introduce the "passenger briefing" and the "engine fire during start brief".

Engine Handling

- Highlight the red line on the RPM gauge.
- Ensure that the student understands the negative effects of "redlining" the engine.
- When stationary on the ground, RPM should be set to 1000 RPM.
- Do not lean the mixture below 3,000 ft.
 - For leaning guidance, refer to the PIM and the leaning guidance on the FIT hub.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Before Takeoff Check

- Have the student run through the checklist.
- Assist if needed.
- Introduce the "takeoff briefing".

Power plant operation

- Introduce that we need to switch the fuel tank every 30 minutes!
- Engine is to be operated at 1000 RPM when stationary on the ground.
- Mixture should be left full rich below 3,000ft.

Primary Effects of Flight Controls

- The initial effect of a flight control is the primary movement it produces when first applied.
 - Ailerons: Control roll about the longitudinal axis (banking the aircraft left or right).
 - Elevator: Controls pitch about the lateral axis (raising or lowering the nose).
 - Rudder: Controls yaw about the vertical axis (moving the nose left or right).
- These immediate responses are the intended primary control movements used to maneuver the aircraft.

Delayed Effects of flight Controls

- The secondary/delayed effect is an additional aerodynamic result produced by the control input after the initial effect.
 - Ailerons: In addition to roll, they can create yaw due to adverse yaw.
 - Elevator: In addition to pitch, it can cause changes in airspeed and lift, which may lead to a climb or descent.
 - Rudder: In addition to yaw, it can cause roll due to the aircraft's dihedral effect.
- Understanding both the initial and secondary effects helps pilots maintain coordinated and controlled flight.

Effects of Airspeed, Slipstream, Trim, Power and Flaps

- Effect of Slipstream
 - Fast:
 - Elevator → More Pitch.
 - Rudder → More Yaw.
 - Ailerons → More Roll.
 - Slow:
 - Elevator → Less Pitch.

- Rudder → Less Yaw.
- Ailerons → Less Roll.
- Effect of Trim
 - Emphasize the importance of trim!
 - Natural sense.
 - Roll wheel forward for forward trim.
 - Roll wheel aft for aft trim.
 - Coarse versus fine trim.
 - Pitch to select attitude, trim to relieve pressure.
 - Any adjustment to pitch or power should be followed with an adjustment to trim.
- Effect of Power and Airspeed
 - For an increase in power.
 - The pitch will increase and there will be a yaw to the left.
 - To correct you need forward pressure on the yoke and right rudder.
 - For a decrease in power.
 - The pitch will decrease and there will be a yaw to the right.
 - To correct you need back pressure on the yoke and left rudder.
 - For an increase in airspeed.
 - The pitch will increase and there will be a yaw to the right.
 - To correct you need forward pressure on the yoke and left rudder.
 - For a decrease in airspeed.
 - The pitch will decrease and there will be a yaw to the left.
 - To correct you need back pressure on the yoke and right rudder.
- Effect of Flaps

Flap Setting / Effects	Lift	Nose	Elevator (initially)	Drag	Airspeed	Elevator (due to drag)
0 - 10 Degrees	↑↑↑	↑↑	Forward	↑	↓	Backward
10 - 25 Degrees	↑↑	↑↑	Forward	↑↑	↓↓	Backward
25 - 40 Degrees	↑	↑	Forward	↑↑↑	↓↓↓	Backward
40 - 25 Degrees	↓	↓	Backward	↓↓↓	↑↑↑	Forward
25 - 10 Degrees	↓↓	↓↓	Backward	↓↓	↑↑	Forward
10 - 0 Degrees	↓↓↓	↓↓	Backward	↓	↑	Forward

Other Controls as Applicable

- Adjusting audio panel power selection, volume, squelch, Garmin430 basic usage (com

- selection, frequency changes, power selection).
- Operation of Mixture Control
 - The mixture lever has a red knob and is located to the right of the throttle lever.
 - Moving the mixture lever forward results in a rich fuel/air mixture, while moving the mixture lever backward results in a lean fuel/air mixture.
 - Emphasize that we do not lean the mixture below 3000 feet.
- Operation of Carburetor Heat
 - The carburetor heat lever is located to the right of the mixture lever.
 - To operate, pull the lever downward to the on position or upward to the off position.
 - Review when carburetor heat is required (show in checklists).
- Operation of Environmental Controls
 - Demonstrate how to operate the cabin heat and defroster

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 105.
- Assign homework:
 - Review primary and secondary flight controls.
 - Have the student memorize the passenger briefing, engine fire during start briefing, and the takeoff briefing.
 - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.

Completion Standards

This lesson is complete when the applicant can:

1. Obtain a weather briefing by using 1-800-WX-BRIEF or 1800wxbrief.com and complete a TOLD card with minimal instructor guidance.
2. Describe the elements of the aircraft's performance charts, weight and balance and environmental factors that affect performance.
3. Demonstrate an understanding of checklist procedures and effects of primary and secondary flight controls with instructor guidance